

Project Initialization and Planning Phase

Date	3 july 2026
Team ID	
Project Title	HDI Prediction System using Machine Learning and Flask
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed HDI Prediction System aims to predict the Human Development Index using machine learning, improving the accuracy and efficiency of socio-economic analysis. It addresses the challenges of manual prediction and data interpretation by providing intelligent insights through predictive analytics and visualization. Key features include an AI-based prediction model, interactive dashboard, and real-time analytical reports.

Project Overview	
Objective	The primary objective is to develop an AI-based Human Development Index prediction system using machine learning techniques, ensuring accurate predictions and supporting effective policy planning.
Scope	The project focuses on collecting and analyzing socio-economic data, training machine learning models, predicting HDI values, and presenting the results through an easy-to-use web application.
Problem Statement	
Description	Current methods of HDI analysis involve manual data processing and statistical interpretation, making prediction time-consuming and less efficient for policy makers and researchers.

Impact	Solving these challenges will improve prediction accuracy, reduce analysis time, support evidence-based policy decisions, and enhance overall socio-economic planning.
Proposed Solution	
Approach	Employ machine learning algorithms to analyze socio-economic indicators and predict Human Development Index values through a scalable and user-friendly web application.
Key Features	- AI-based Human Development Index prediction model. - Interactive dashboard for visualization and analysis. - Accurate prediction using socio-economic indicators. - Fast and user-friendly web application. - Real-time prediction results and graphical reports.



Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel core i5
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask

Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	UNDP/World Bank HDI Dataset,CSV Format